Rishav Saigal

Senior Data Scientist | Machine Learning Engineer | Generative AI / Agentic AI Engineer | AI Architect

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PROFESSIONAL SUMMARY

Senior Data Scientist and ML/GenAI Engineer with **8+ years** of experience architecting and deploying **enterprise-grade, production-ready AI/ML systems** across healthcare, supply chain, and media verticals. Deep expertise in **Generative AI (GenAI)**, **Agentic AI** (Google ADK, LangGraph, CrewAI), **Retrieval-Augmented Generation (RAG)**, **Prompt Engineering**, **Large Language Model (LLM) applications**, and **Time Series MLOps**. Experienced in production AI architecture, autonomous ML orchestration, multi-agent systems, and enterprise AI deployment. Author of **11 independent technical articles** (Towards AI, Towards Deep Learning, Medium, LinkedIn) plus three open, DOI-indexed **research software & dataset releases** (Zenodo) on LLM behavioral drift and multi-axis model routing, a self-coined **calcification effect**, multi-agent robustness, autonomous ML, and AI agent production economics. Active open-source contributor (PyCaretAgent, CognitoEDA, LLM Drift Experiment, model-router, a2ui-ad-component).

TECHNICAL SKILLS

- **Generative AI & LLMs:** Generative AI (GenAI), Large Language Models (LLMs), Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, CrewAI, Google ADK, FastMCP, Hugging Face, Ollama
- **Machine Learning & Statistical Modeling:** Scikit-Learn, TensorFlow, Keras, PyCaret, SHAP / Explainable AI (XAI), Transfer Learning, Survival Analysis, Causal Inference, Time Series Forecasting, Hierarchical Linear Models (HLM)
- **MLOps & Data Engineering:** MLflow, DataRobot, Azure Databricks, Docker, CI/CD for ML, Model Monitoring, Automated Retraining, Feature Engineering, PySpark, Dask, Pandas, NumPy, SciPy
- **Cloud & AI Platforms:** AWS (SageMaker, Lambda, ECS, Transcribe, Comprehend), Azure ML, GCP (Vertex AI, Cloud Run, Gemini Enterprise, Agent Engine Platform)
- **Programming, APIs & Databases:** Python, JavaScript, FastAPI, Flask, REST APIs, MSSQL, MySQL, PostgreSQL, Redis
- **Visualization & Applications:** Streamlit, DASH, Plotly, Seaborn, Matplotlib, Power BI, ipywidgets
- **Leadership & Architecture:** Team Leadership, Agile/Scrum, Technical Roadmapping, Stakeholder Management, Effort Estimation, Architectural Design, Proof of Concept (POC) Management, Impact & Risk Assessment, Cost Optimisation

PROFESSIONAL EXPERIENCE

Cognizant | *Kolkata*

Manager (Senior Data Scientist / AI Architect) | *Nov 2023 – Present*

- **Enterprise AI Architecture & Strategy (SME):** Served as designated Subject Matter Expert (SME) in AI Architecture Design and GenAI; spearheaded end-to-end architectural roadmaps and technical standards for enterprise-grade Agentic AI systems across complex GCP infrastructure (Vertex AI, Cloud Run, Gemini Enterprise, Agent Engine Platform, FastMCP); designed autonomous multi-agent orchestration frameworks and token-cost observability.
- **RFP Solutioning, Pre-Sales & Resource Planning:** Spearheaded architectural solution design and technical response strategy for multiple high-value RFPs as a core SME; formulated tailored architecture proposals, multi-phase resource planning models, and effort estimation matrices for enterprise executive buyers.
- **Client Deliverable Management & Engagement:** Led end-to-end delivery ownership from initial client requirement discovery workshops through technical architecture, executive stakeholder POC demonstrations, and production sign-off governance across healthcare, media, and enterprise domains.
- **Team Leadership, Mentorship & Talent Acquisition:** Managed and mentored cross-functional teams of ML/GenAI engineers; directed campus-to-corporate mentorship programmes for college freshers and actively contributed to tech communities fostering student growth; served as primary technical panellist conducting hiring interviews for account staffing and external lateral recruitment.
- **Production AI Governance & Reliability:** Enforced robust code review standards, CI/CD automated retraining workflows, and responsible AI guardrails; implemented self-correction validation loops, intent-routing architectures, and enterprise-grade reliability benchmarks across production pipelines.

Genpact | *Bengaluru*

Assistant Manager | *Jun 2022 – Nov 2023*

- **Team Leadership & ML Architecture:** Managed and mentored a cross-functional team of 6 ML engineers; led Agile architecture reviews, sprint delivery, and end-to-end MLOps solution deployments on AWS SageMaker and cloud environments; defined engineering best practices and delivery roadmaps.
- **Executive Stakeholder Delivery:** Served as primary technical point-of-contact for executive client demos and quarterly business reviews; translated commercial objectives into scalable machine learning specifications, delivering multi-million-dollar projected operational cost savings at 90%+ model accuracy.
- **Responsible AI Governance & Explainability:** Architected production model interpretability frameworks using SHAP (XAI), customer segmentation, and causal inference uplift modeling; upskilled client data science teams on responsible AI governance and proactive drift monitoring.

Capgemini | *Bengaluru*

Consultant | *Apr 2018 – Jun 2022*

- **Technical Lead & Full-Lifecycle Delivery:** Led cross-functional engineering teams of up to **15 members** across supply chain, manufacturing, and enterprise service domains; directed end-to-end ML solution delivery from Agile discovery and architecture through production deployment on AWS and Azure cloud infrastructure.
- **Distributed Time Series & MLOps Architecture:** Engineered scalable distributed time series and predictive maintenance pipelines on Azure Databricks and PySpark; automated feature engineering, hyperparameter tuning, distributed model selection, and continuous retraining workflows, cutting training latency from days to hours.
- **Deep Learning & Computer Vision Systems:** Architected sequence volume forecasting architectures (LSTM, MLP) with Hungarian dispatch heuristics and frame-level CNN transfer learning pipelines for real-time asset detection; delivered interactive Plotly operational HUDs for executive decision-making.

EDUCATION

Indian Institute of Technology Patna — Ph.D. — Computer Science & Engineering (Pursuing)
Jul 2026 – Jun 2029 (Expected)

Birla Institute of Technology and Science, Pilani — M.Tech — Data Science & Engineering
Sep 2019 – Aug 2021

RCC Institute of Information Technology (MAKAUT) — B.Tech — Computer Science & Engineering
Aug 2013 – Jul 2017

RESEARCH INTERESTS

LLM behavioral drift, adversarial robustness, and a self-coined calcification effect in multi-agent debate systems; autonomous ML orchestration and agentic pipeline design (AutoML, self-correcting agents); probabilistic time series forecasting with LLMs; production economics and efficiency optimisation of large-scale AI agent deployments.

PUBLICATIONS & INDEPENDENT RESEARCH

The $300B Search Dilemma: Decoupling Commercial Components from LLM Reasoning in Conversational AI — *Towards Deep Learning* | Nov 2026 | **[Article] [Repo] [DOI]**

The Specialized Frontier: An Inquiry into Gated AI Architectures and the Cooperative Safety Flywheel — *Towards AI* | Aug 2026 | **[Article]**

Claude Model Routing: Stop Scoring "Complexity." Score These Two Things Instead. — *Towards Deep Learning* | Jul 2026 | **[Article] [DOI]**

LangGraph Multi-Agent Architecture: Building a Self-Critiquing AI Debate System — *Towards AI* | May 2026 | **[Article]**

LLM Drift Experiment: A Quantitative Framework for Measuring Behavioural Drift via Adversarial Multi-Agent Debate — *Zenodo* | May 2026 | **[Research Dataset & Framework | DOI]**

Measuring Behavioral Drift in LLMs: 22 Signals, 5 Dimensions, and the Calcification Effect — *Towards AI* | May 2026 | **[Article]**

LLM Drift Explained: Do AI Models Lose Themselves Under Adversarial Pressure? — *Towards AI* | May 2026 | **[Article]**

The $1.5 Million Difference: Why Benchmarks are Only 10% of the AI Agent Story — *LinkedIn* | Mar 2026 | **[Article]**

AutoML on Autopilot — *Towards AI* | Mar 2026 | **[Article]**

CognitoEDA — *Medium* | Aug 2025 | **[Article]**

Beyond Prediction: Generative AI's Probabilistic Future in Time Series — *Medium* | Jun 2025 | **[Article]**

Unlocking Time Series with LLMs: A New Era with TimeCAP — *Medium* | Jun 2025 | **[Article]**

PROJECTS

Generative AI (GenAI) Video Generation *Jan 2026 – Present*
Autonomous multi-agent pipeline using Google ADK, GCP Vertex AI (document storage), Gemini Enterprise, Agent Engine Platform, and FastMCP; features context-window summarisation, token-cost observability, dynamic prompting, and automated self-correction validation loops.
Impact: Slashed end-to-end delivery timeline from 5–14 days down to **under 6 hours** (from script generation to final video demo).

Media Analytics (POC) *Jun 2025 – Oct 2025*
Predictive ML and prompt engineering pipeline for inference explanation, insight extraction, and business KPI generation; models multi-channel budget allocation and predicts future investment growth using multi-timeseries forecasting.

Price Elasticity (POC) *Jun 2025 – Jul 2025*
Dynamic pricing architecture using Hierarchical Linear Models (HLM) (statsmodels, Python, Azure ML) with automated MLflow continuous retraining pipelines to model cross-product elasticity frontiers.

Enterprise Analytics Chatbot *Jul 2024 – Dec 2025*
Conversational analytics agent (Streamlit, PostgreSQL) featuring a dedicated Intent Classifier to determine required insight types, paired with dynamic Text-to-SQL generation powered by Client Proprietary Internal LLMs.
Impact: Projected cycle time reduction for designing complex enterprise insights from 3 months down to **under 15 days**.

Intelligent QA / BA Chatbot *Apr 2024 – Apr 2025*
Enterprise conversational system (FastAPI, Python) utilizing semantic policy retrieval, intent classifiers, and structured intent flows for automated policy Q&A and operational task execution.
Impact: Autonomously resolved **80% of internal policy document queries**; automated RBAC-governed workflow execution and access request approvals/rejections.

LLM-Driven Predictive Modelling *Jul 2024 – Sep 2024*
Unsupervised feature extraction pipeline using open-source LLMs via Ollama to discover ticket objective classes; aggregates class occurrences into hourly instance feature vectors fed into an existing deep LSTM sequence model for enhanced predictive sequence accuracy.

LLM-Based Test Case Generator *Nov 2023 – Mar 2024*
QA automation toolchain (FastAPI, Python) parsing unstructured Business Requirement Documents (BRDs) to extract acceptance criteria and auto-generate structured Gherkin and TMMi test suites.
Impact: Accelerated test authoring cycle by **70%**; ensured 100% test-to-requirement coverage mapping.

Healthcare Consumer Analytics *Jun 2022 – Oct 2023*
Production ML pipeline on AWS SageMaker (Scikit-Learn, DataRobot) combining customer segmentation, causal inference uplift modeling, and SHAP explainability for account-level promotional strategy ranking.
Impact: Delivered projected **$2.3M in cost savings** at **90%+ model accuracy** across all account-level promotion campaigns.

Demand Sensing MLOps *Mar 2020 – Jun 2022*
Distributed time series MLOps pipeline on Azure Databricks (PySpark, TensorFlow, Keras) with automated feature engineering, distributed model selection, and continuous retraining workflows.
Impact: Achieved **5% accuracy improvement**, **10% better model selection**, reduced training from 3 days to **2 hours**, and **60% reduction** in manual intervention.

Predictive Maintenance *Dec 2019 – Mar 2020*
Time-varying survival analysis pipeline (Cox Proportional Hazards, XGBoost, lifelines, Python) predicting EV component degradation windows paired with interactive Plotly operational HUDs.
Impact: Achieved **82% validation accuracy** for early failure prevention.

Service Now Ticket Analytics *Oct 2019 – Dec 2019*
Deep learning sequence volume forecasting architecture (LSTM vs. MLP, TensorFlow, MySQL) featuring custom greedy Hungarian workload dispatch heuristics for agent routing.
Impact: Achieved **85% validation accuracy** with intelligent routing heuristics for optimized workload dispatch.

Supply Chain Demand Forecasting (POC) *Apr 2019 – Oct 2019*
Multivariate demand forecasting pipeline (statsmodels SARIMAX, pandas, Python) integrating dynamic NLTK VADER social media sentiment indices as exogenous regressors with grid-search hyperparameter optimization.

Retail Video Analytics (POC) *Apr 2018 – Mar 2019*
Computer vision pipeline (OpenCV, TensorFlow, Keras, Python) featuring frame-level CNN transfer learning for brand asset detection and multi-class classification across retail video streams.

CERTIFICATIONS

- **Generative AI Practitioner Certificate** — Dataiku Academy, Jun 2026. **[Credential]**
- **Microsoft Certified: Azure AI Fundamentals** — Microsoft, Jun 2025. **[Credential]**
- **MLOps Practitioner Certificate** — Dataiku Academy, Aug 2023. **[Credential]**
- **Advanced Designer Certificate** — Dataiku Academy, Jul 2023. **[Credential]**
- **ML Practitioner Certificate** — Dataiku Academy, Jul 2023. **[Credential]**
- **Core Designer Certificate** — Dataiku Academy, Jul 2023. **[Credential]**